# Laksh Goyal

San Diego, CA, US | +1 (858) 298-8940 | lgoyal@ucsd.edu | lakshgoyal.com | linkedin.com/in/lakshgoyal | github.com/lgoyal6

## Education

**University of California San Diego** GPA: 3.6

*B.S. in Computer Science; Minors in Cognitive Science and Business Economics* *Expected Graduation: Dec 2027*

**Relevant Coursework**: Algorithms & Data Structures; Operating Systems; Computer Architecture

## Experience

**DR Financial** Remote

*Software Engineer Intern* *Jul 2026 – Present*

- Built DR Financial's first product for checking doctors' licenses and credentials, replacing 4 hours of manual lookups per case with 2 minutes of automated checks and 10 minutes of human review.
- Detected every excluded doctor across 1,500 adversarially sampled cases against the 83,665-row federal exclusion list, establishing a 95% confidence upper bound under 0.20% for missed exclusions
- Shipped the credentialing service on AWS ECS/Fargate, using ECR for Docker images and S3 for reviewer evidence.

**GMI Cloud** (**NVIDIA** Reference Platform Cloud Partner) Mountain View, CA

*Software Engineer Intern* *Jun 2026 – Sep 2026*

- Cut inference cost 47% by routing coding requests to a cheaper model first, escalating only when an automated checker rejected output, and scoring 94/100 on EvalPlus across 500+ requests on the frozen production evaluation set
- Built Mint, a React/FastAPI platform for sales-lead sourcing, qualification and HubSpot handoff, cutting each batch from 5 people at 4 hours each to 20 minutes of human review by 1 person.
- Gated Mint's React/FastAPI/PostgreSQL releases with Product and Operations, catching 15 Playwright regressions.

**Qualcomm Institute** San Diego, CA

*Software Engineer, ML Systems (Part-time)* *Oct 2025 – Jun 2026*

- More than doubled training-ready wildlife audio, recovering 15% of raw recordings versus 7% before, by validating and filtering corrupted files across multi-terabyte archives for EfficientNet
- Increased wildlife-audio training throughput 30% with prefetching and parallel data loading, cutting GPU time waiting for data from 31% to 10% while feeding EfficientNet training runs.

**AWS Student Builder Club @ UC San Diego** | Cloud Engineering Lead San Diego, CA

- Cut Docker and AWS setup to 30 minutes for 20 club members with hands-on workshops and a shared installation runbook.

## Achievements

**11x Hackathon Winner**: 1st Akash, 2nd Replay (both AWS), 2nd Bow Capital Defense, 1st Convex, 1st Runloop/Reflex

**2x Founding Engineer**: Pylon ($100K raised, Bow Capital incubator); Inovasity (3D wound imaging and analysis)

**Open Source Contributor**: Awkward Array, https://github.com/scikit-hep/awkward (1K+ GitHub stars)

## Projects

**Strata**: **LSM-Tree Storage Engine** | C++20 Demo | github.com/lgoyal6/strata

- Implemented a durable C++20 key-value engine that beat RocksDB 1.24× on a 1M-record load and 50× on synchronous commits by batching concurrent writes through group commit in published macOS benchmarks
- Cut internal write overhead 26% and compaction time 48% with two background workers across 3M updates

**Basis**: **Brokerage Ledger** | Java 21, Spring Boot Live | github.com/lgoyal6/basis

- Avoided $1,200 in personal brokerage fees with Basis, a Java double-entry ledger that reconstructs holdings from trades and traces cost-basis discrepancies to the underlying transactions.
- Serialized writes per account with a PostgreSQL advisory lock to prevent duplicate share sales; matched race tests failed without the lock and passed with it under identical test inputs

**GPU Broker**: **GPU Job Queue** | Python, AWS Demo | github.com/lgoyal6/gpu-broker

- Ran a GPU job broker used by 20 AWS Student Builder Club members, including 8 repeat users, putting paid cloud, spot instances and a shared lab A6000 behind one queue, with per-user dollar budgets governing every job
- Injected 315 spot interruptions across 100 concurrent training jobs, checkpointing and resuming each job and completing all 100 exactly once against a matched uninterrupted control run

**Hacklist**: **Hackathon Calendar** | JavaScript Live | github.com/lgoyal6/hacklist

- Expanded Hacklist to San Francisco and San Diego with separate calendar feeds and reserved crawl capacity, reaching 100+ SF users and 50 SD users while protecting local search slots from SF's larger source graph
- Tracked impressions and clicks with rotating anonymous IDs, rejected fields that could contain personal information, and kept recommendations working with a fixed fallback when the learned ranker was unavailable

## Technical Skills

**Languages**: Python, C++20, Java 21, Go, Rust, C, TypeScript, JavaScript, SQL, Bash, HTML/CSS, Swift, Kotlin, R

**AI & Backend**: PyTorch, Transformers, scikit-learn, NumPy, FastAPI, Spring Boot, Flask, Node.js, React, REST APIs, gRPC

**Data & Infrastructure**: PostgreSQL, Redis, SQLite, Docker, Kubernetes, AWS EKS, EC2, Lambda, S3, IAM, Terraform

**Systems & Tooling**: Linux, Git, GitHub Actions, CI/CD, CMake, Gradle, pytest, JUnit 5, Playwright, libFuzzer, Prometheus